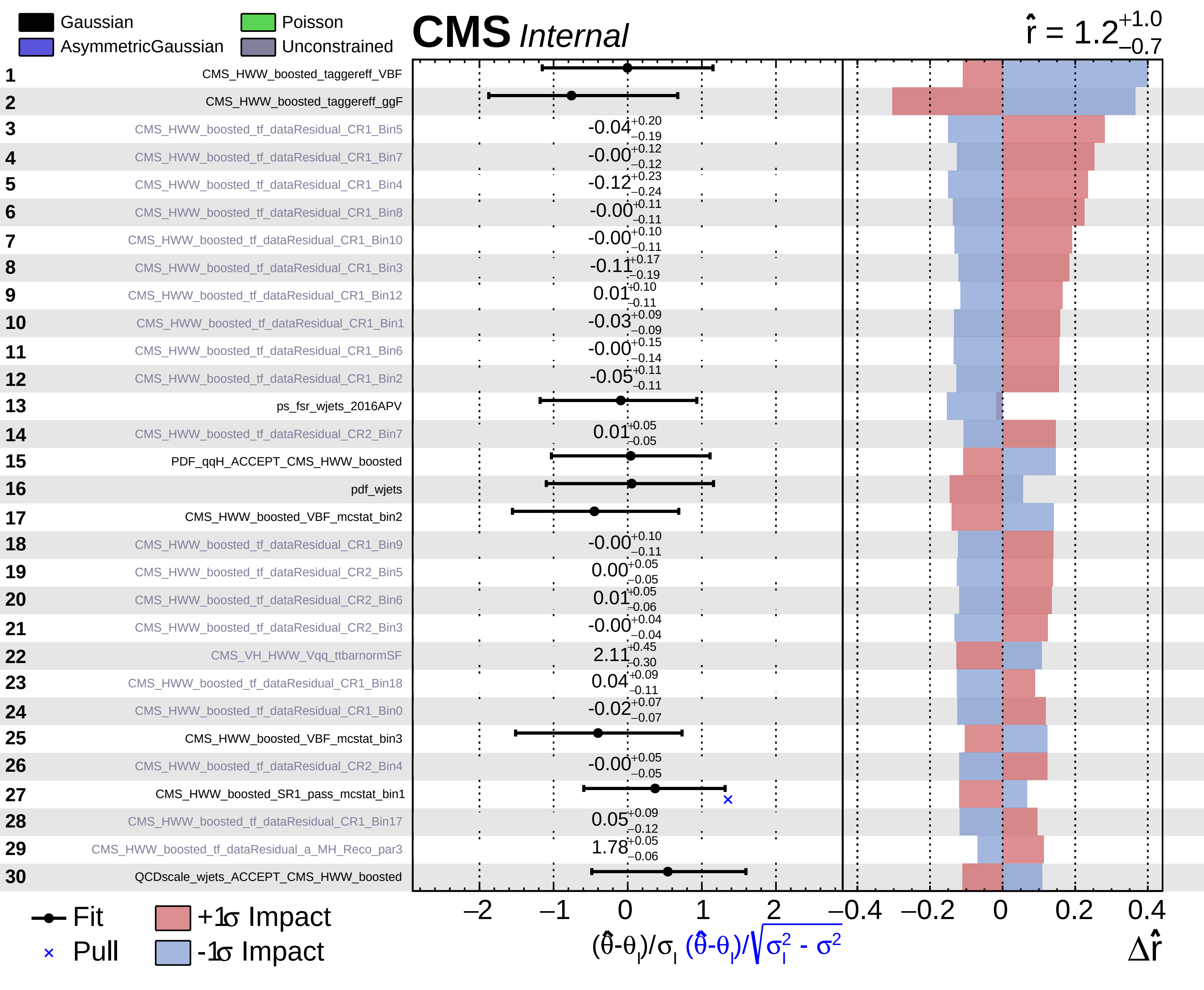
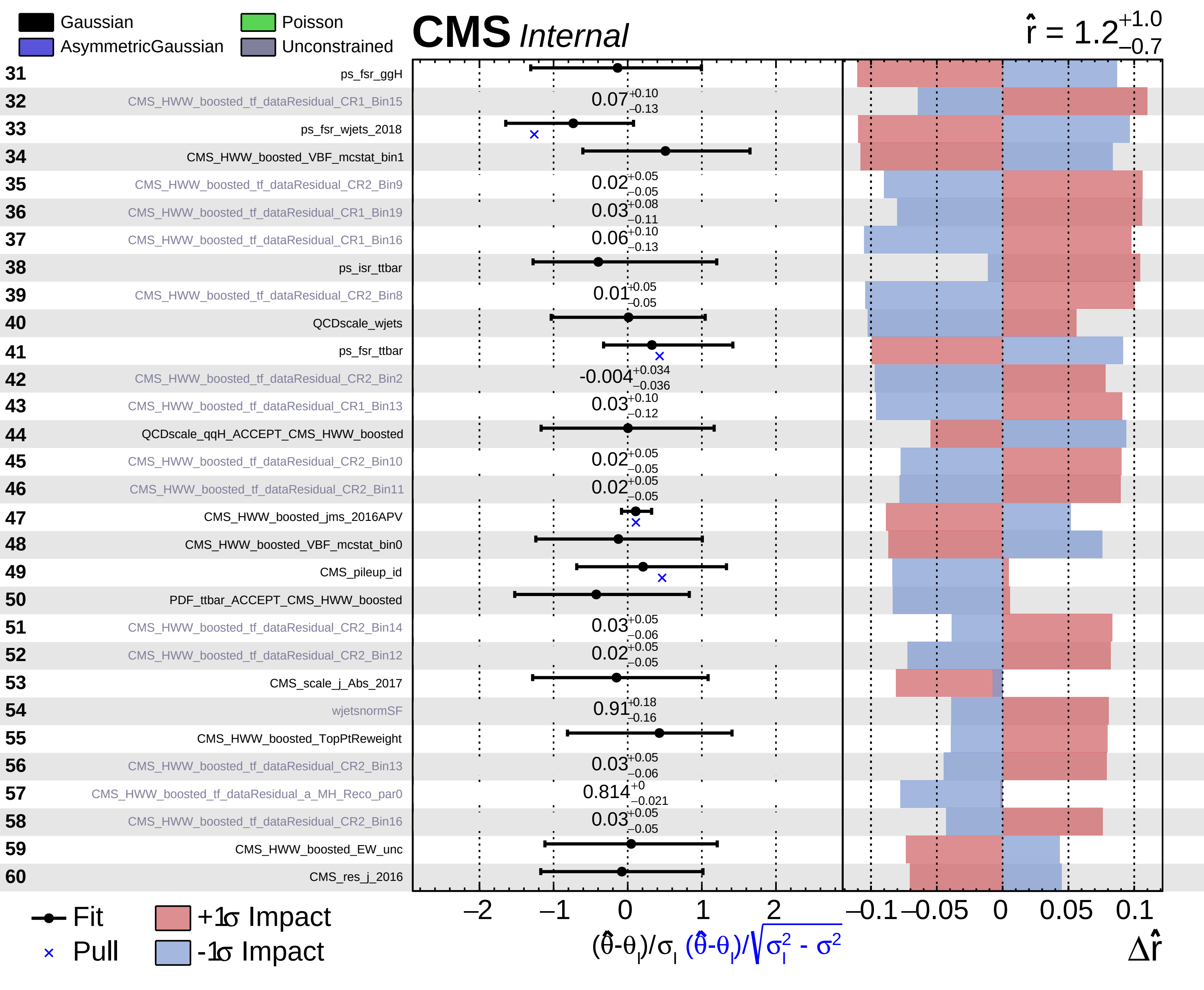
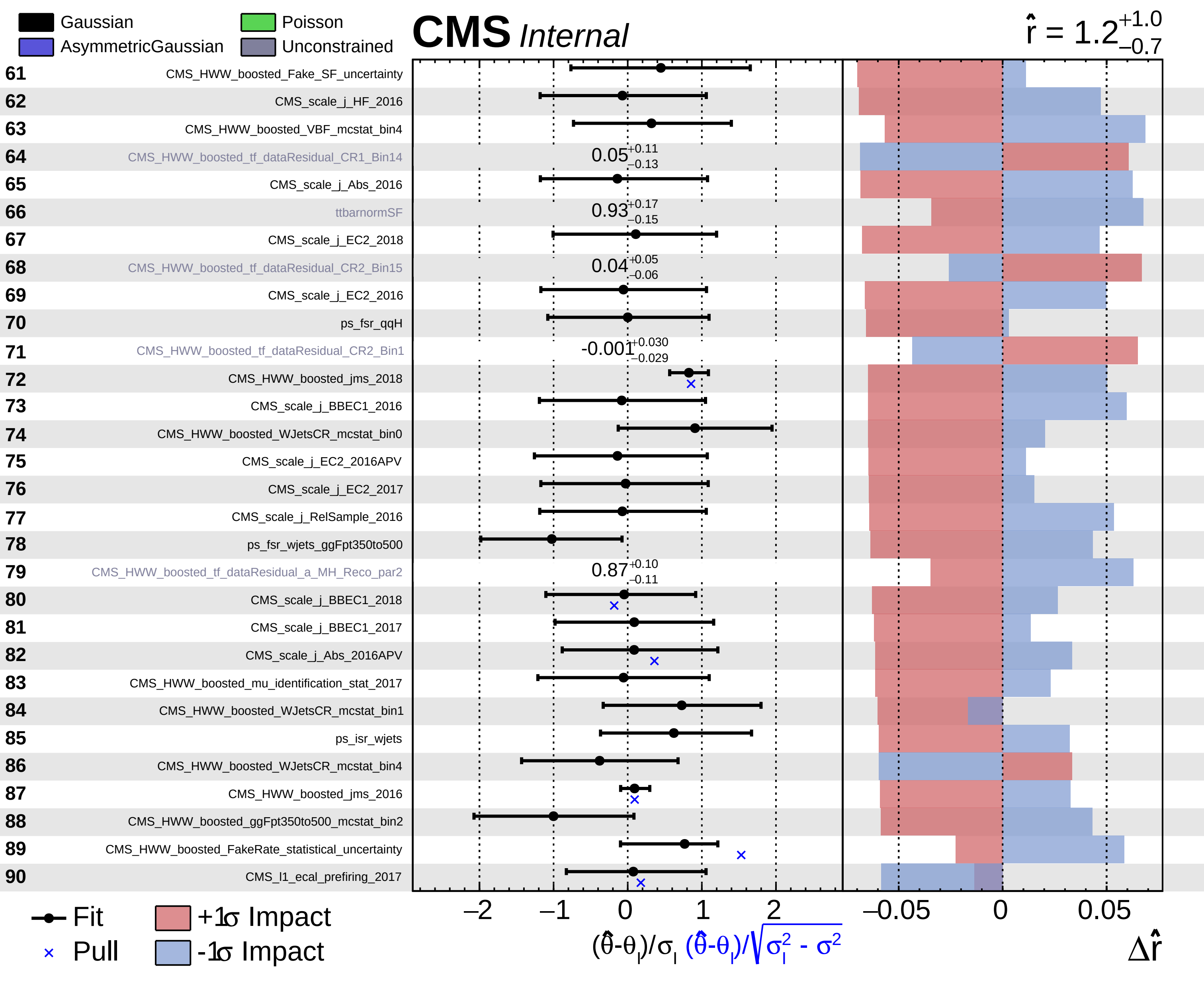
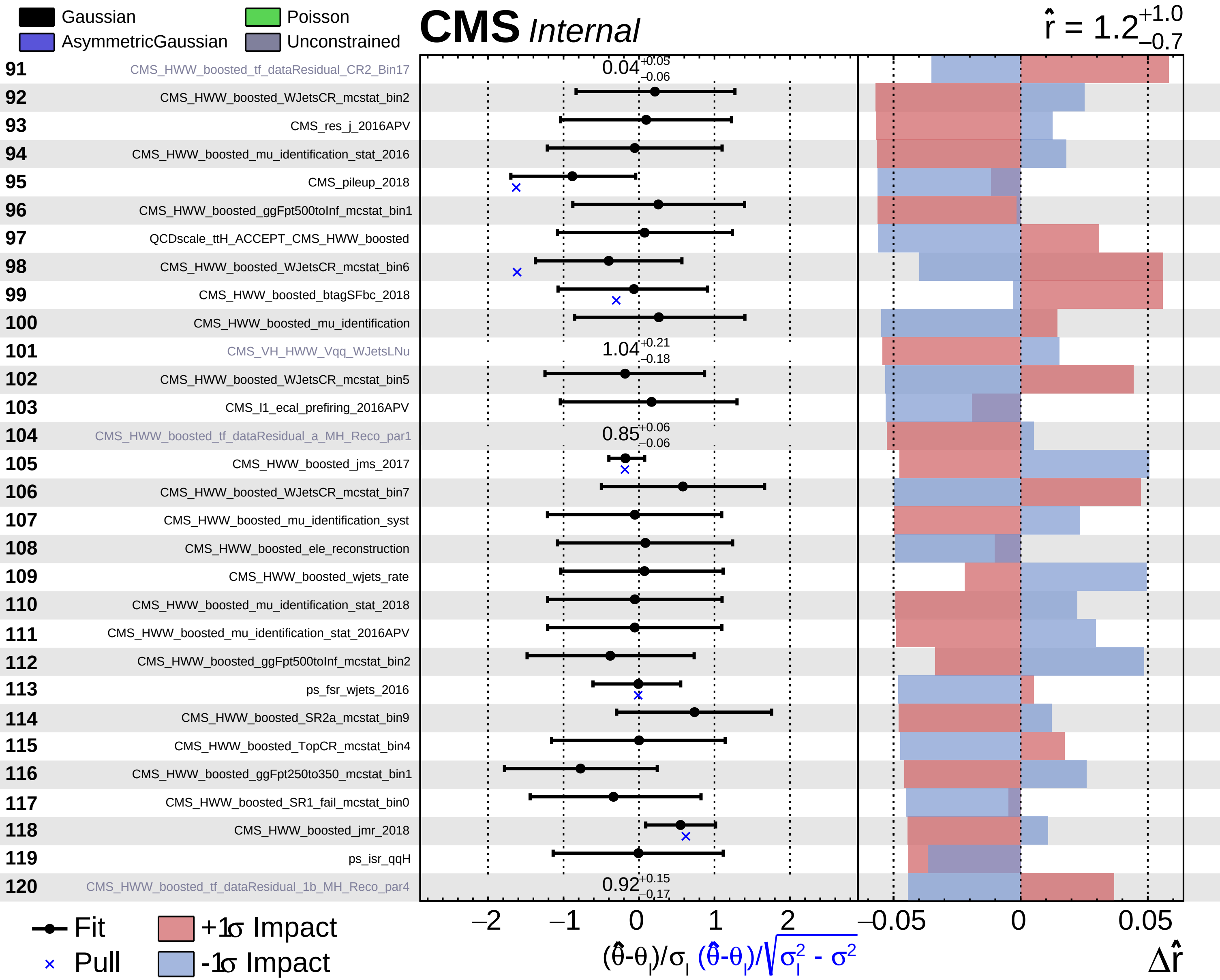

Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
r̂ = 1.2 +1.0 −0.7
1 CMS_HWW_boosted_taggereff_VBF
2 CMS_HWW_boosted_taggereff_ggF
3 CMS_HWW_boosted_tf_dataResidual_CR1_Bin5 -0.04 +0.20 −0.19
4 CMS_HWW_boosted_tf_dataResidual_CR1_Bin7 -0.00 +0.12 −0.12
5 CMS_HWW_boosted_tf_dataResidual_CR1_Bin4 -0.12 +0.23 −0.24
6 CMS_HWW_boosted_tf_dataResidual_CR1_Bin8 -0.00 +0.11 −0.11
7 CMS_HWW_boosted_tf_dataResidual_CR1_Bin10 -0.00 +0.10 −0.11
8 CMS_HWW_boosted_tf_dataResidual_CR1_Bin3 -0.11 +0.17 −0.19
9 CMS_HWW_boosted_tf_dataResidual_CR1_Bin12 0.01 +0.10 −0.11
10 CMS_HWW_boosted_tf_dataResidual_CR1_Bin1 -0.03 +0.09 −0.09
11 CMS_HWW_boosted_tf_dataResidual_CR1_Bin6 -0.00 +0.15 −0.14
12 CMS_HWW_boosted_tf_dataResidual_CR1_Bin2 -0.05 +0.11 −0.11
13 ps_fsr_wjets_2016APV
14 CMS_HWW_boosted_tf_dataResidual_CR2_Bin7 0.01 +0.05 −0.05
15 PDF_qqH_ACCEPT_CMS_HWW_boosted
16 pdf_wjets
17 CMS_HWW_boosted_VBF_mcstat_bin2
18 CMS_HWW_boosted_tf_dataResidual_CR1_Bin9 -0.00 +0.10 −0.11
19 CMS_HWW_boosted_tf_dataResidual_CR2_Bin5 0.00 +0.05 −0.05
20 CMS_HWW_boosted_tf_dataResidual_CR2_Bin6 0.01 +0.05 −0.06
21 CMS_HWW_boosted_tf_dataResidual_CR2_Bin3 -0.00 +0.04 −0.04
22 CMS_VH_HWW_Vqq_ttbarnormSF 2.11 +0.45 −0.30
23 CMS_HWW_boosted_tf_dataResidual_CR1_Bin18 0.04 +0.09 −0.11
24 CMS_HWW_boosted_tf_dataResidual_CR1_Bin0 -0.02 +0.07 −0.07
25 CMS_HWW_boosted_VBF_mcstat_bin3
26 CMS_HWW_boosted_tf_dataResidual_CR2_Bin4 -0.00 +0.05 −0.05
27 CMS_HWW_boosted_SR1_pass_mcstat_bin1
28 CMS_HWW_boosted_tf_dataResidual_CR1_Bin17 0.05 +0.09 −0.12
29 CMS_HWW_boosted_tf_dataResidual_a_MH_Reco_par3 1.78 +0.05 −0.06
30 QCDscale_wjets_ACCEPT_CMS_HWW_boosted
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
Fit
Pull
+1σ Impact
-1σ Impact
(θ̂-θI)/σI (θ̂-θI)/√(σI² - σ²)
Δr̂

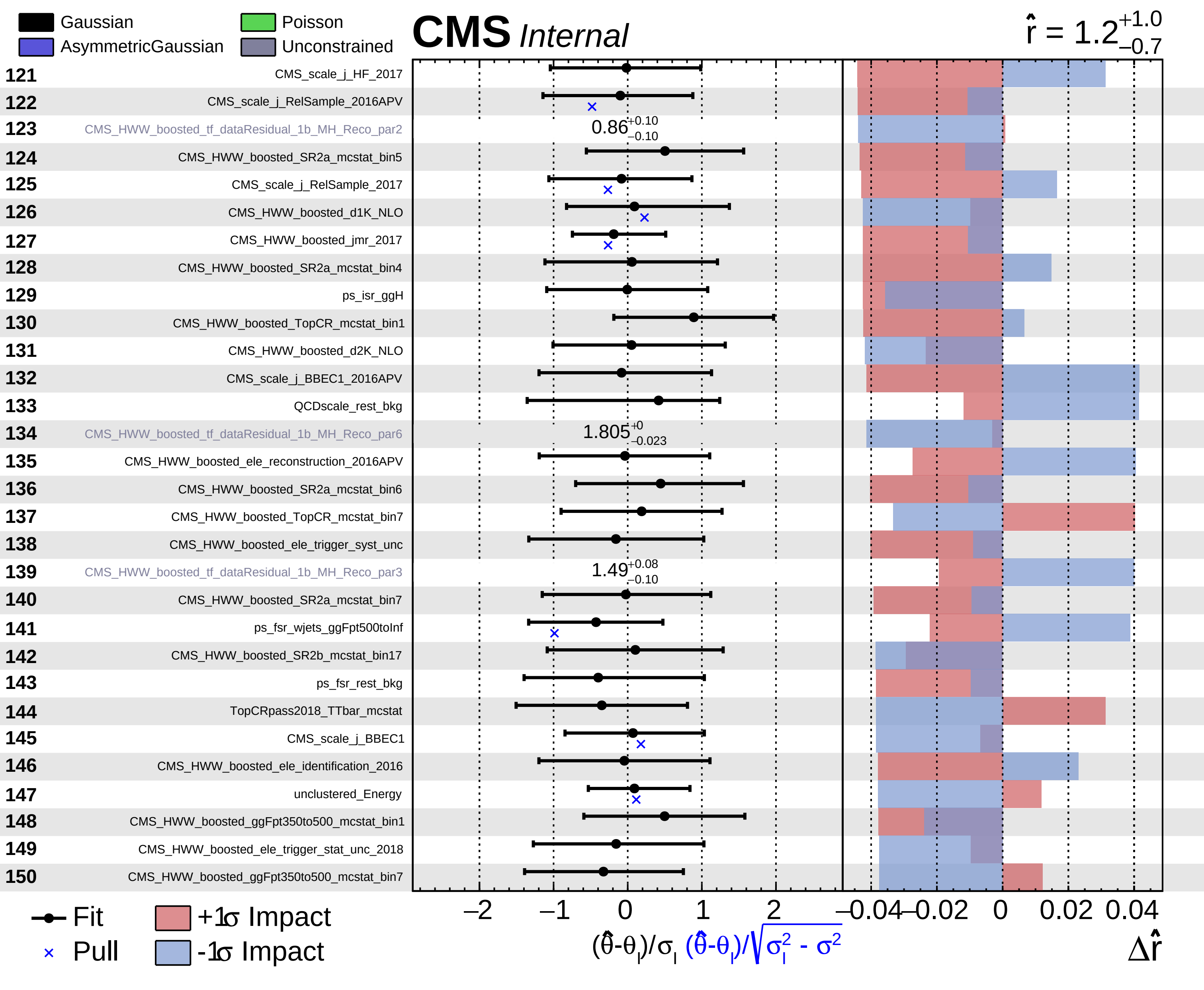

Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.2^{+1.0}_{-0.7}$
31 ps_fsr_ggH
32 CMS_HWW_boosted_tf_dataResidual_CR1_Bin15
$0.07^{+0.10}_{-0.13}$
33 ps_fsr_wjets_2018
34 CMS_HWW_boosted_VBF_mcstat_bin1
35 CMS_HWW_boosted_tf_dataResidual_CR2_Bin9
$0.02^{+0.05}_{-0.05}$
36 CMS_HWW_boosted_tf_dataResidual_CR1_Bin19
$0.03^{+0.08}_{-0.11}$
37 CMS_HWW_boosted_tf_dataResidual_CR1_Bin16
$0.06^{+0.10}_{-0.13}$
38 ps_isr_ttbar
39 CMS_HWW_boosted_tf_dataResidual_CR2_Bin8
$0.01^{+0.05}_{-0.05}$
40 QCDscale_wjets
41 ps_fsr_ttbar
42 CMS_HWW_boosted_tf_dataResidual_CR2_Bin2
$-0.004^{+0.034}_{-0.036}$
43 CMS_HWW_boosted_tf_dataResidual_CR1_Bin13
$0.03^{+0.10}_{-0.12}$
44 QCDscale_qqH_ACCEPT_CMS_HWW_boosted
45 CMS_HWW_boosted_tf_dataResidual_CR2_Bin10
$0.02^{+0.05}_{-0.05}$
46 CMS_HWW_boosted_tf_dataResidual_CR2_Bin11
$0.02^{+0.05}_{-0.05}$
47 CMS_HWW_boosted_jms_2016APV
48 CMS_HWW_boosted_VBF_mcstat_bin0
49 CMS_pileup_id
50 PDF_ttbar_ACCEPT_CMS_HWW_boosted
51 CMS_HWW_boosted_tf_dataResidual_CR2_Bin14
$0.03^{+0.05}_{-0.06}$
52 CMS_HWW_boosted_tf_dataResidual_CR2_Bin12
$0.02^{+0.05}_{-0.05}$
53 CMS_scale_j_Abs_2017
54 wjetsnormSF
$0.91^{+0.18}_{-0.16}$
55 CMS_HWW_boosted_TopPtReweight
56 CMS_HWW_boosted_tf_dataResidual_CR2_Bin13
$0.03^{+0.05}_{-0.06}$
57 CMS_HWW_boosted_tf_dataResidual_a_MH_Reco_par0
$0.814^{+0}_{-0.021}$
58 CMS_HWW_boosted_tf_dataResidual_CR2_Bin16
$0.03^{+0.05}_{-0.05}$
59 CMS_HWW_boosted_EW_unc
60 CMS_res_j_2016
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



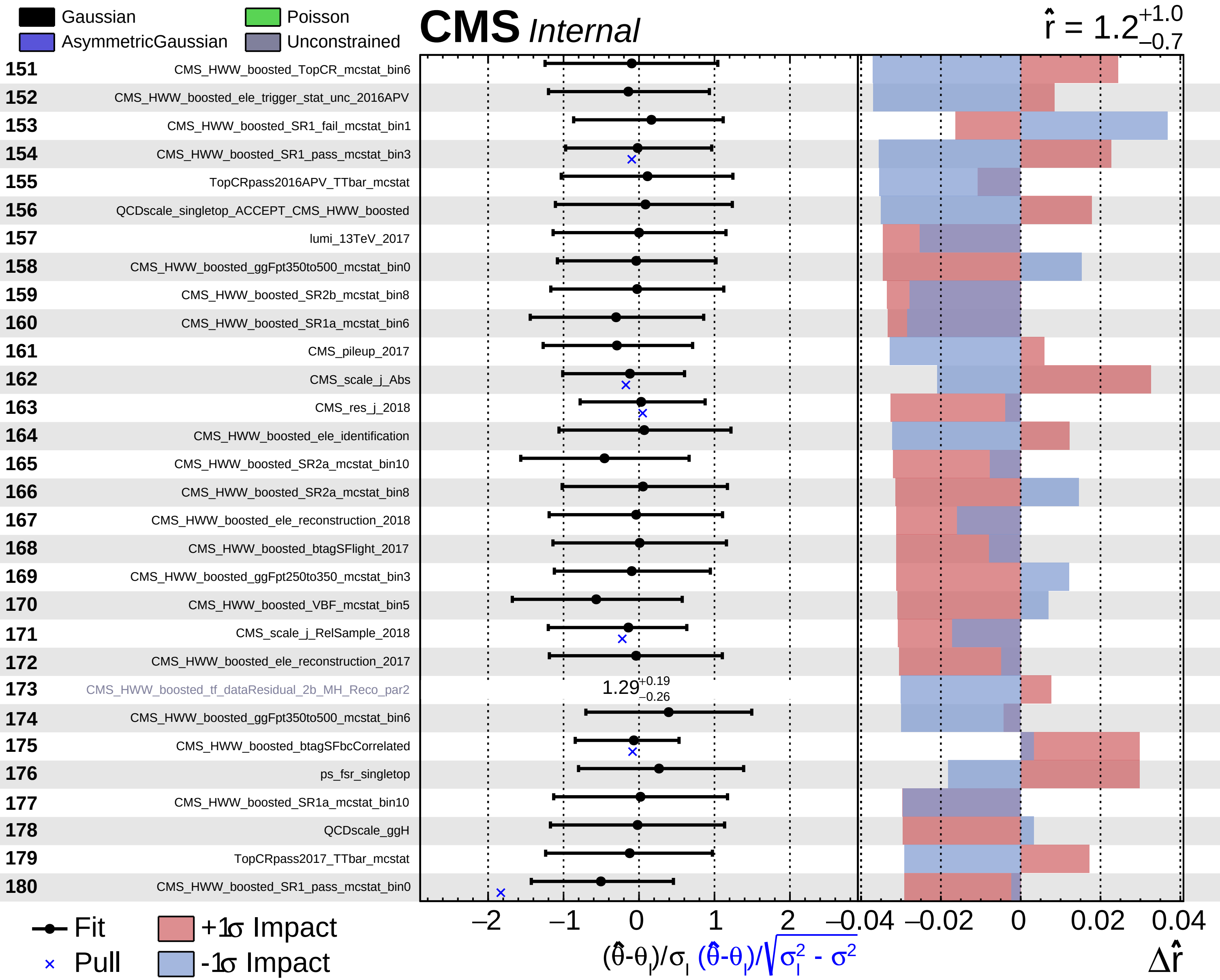
Gaussian
AsymmetricGaussian
Poisson
Unconstrained
CMS Internal
r̂ = 1.2+1.0 −0.7
61 CMS_HWW_boosted_Fake_SF_uncertainty
62 CMS_scale_j_HF_2016
63 CMS_HWW_boosted_VBF_mcstat_bin4
64 CMS_HWW_boosted_tf_dataResidual_CR1_Bin14
0.05+0.11 −0.13
65 CMS_scale_j_Abs_2016
66 ttbarnormSF
0.93+0.17 −0.15
67 CMS_scale_j_EC2_2018
68 CMS_HWW_boosted_tf_dataResidual_CR2_Bin15
0.04+0.05 −0.06
69 CMS_scale_j_EC2_2016
70 ps_fsr_qqH
71 CMS_HWW_boosted_tf_dataResidual_CR2_Bin1
-0.001+0.030 −0.029
72 CMS_HWW_boosted_jms_2018
73 CMS_scale_j_BBEC1_2016
74 CMS_HWW_boosted_WJetsCR_mcstat_bin0
75 CMS_scale_j_EC2_2016APV
76 CMS_scale_j_EC2_2017
77 CMS_scale_j_RelSample_2016
78 ps_fsr_wjets_ggFpt350to500
79 CMS_HWW_boosted_tf_dataResidual_a_MH_Reco_par2
0.87+0.10 −0.11
80 CMS_scale_j_BBEC1_2018
81 CMS_scale_j_BBEC1_2017
82 CMS_scale_j_Abs_2016APV
83 CMS_HWW_boosted_mu_identification_stat_2017
84 CMS_HWW_boosted_WJetsCR_mcstat_bin1
85 ps_isr_wjets
86 CMS_HWW_boosted_WJetsCR_mcstat_bin4
87 CMS_HWW_boosted_jms_2016
88 CMS_HWW_boosted_ggFpt350to500_mcstat_bin2
89 CMS_HWW_boosted_FakeRate_statistical_uncertainty
90 CMS_l1_ecal_prefiring_2017
−2 −1 0 1 2
−0.05 0 0.05
Fit
Pull
+1σ Impact
−1σ Impact
(θ̂-θI)/σI (θ̂-θI)/√(σI² - σ²)
Δr̂

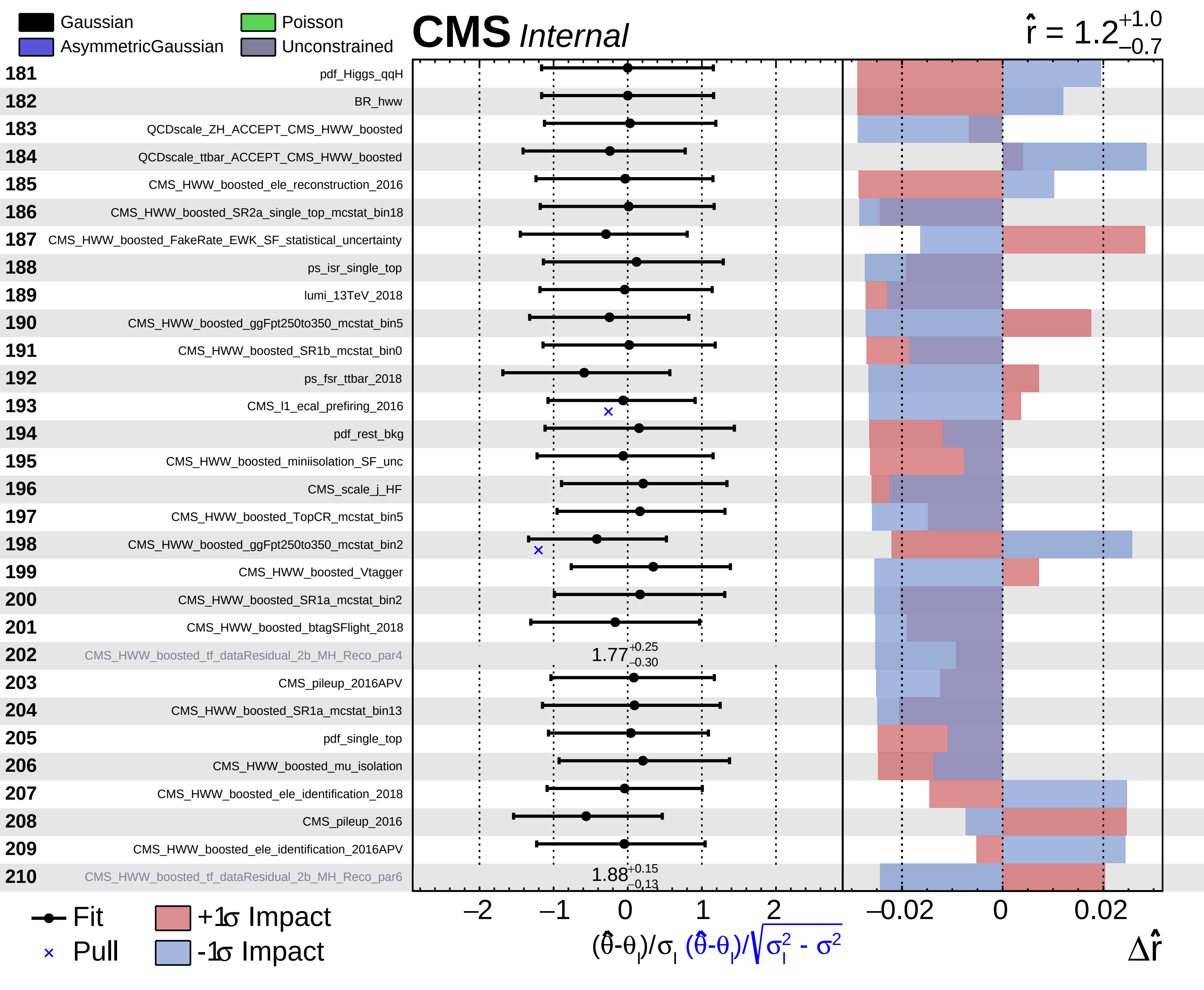


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.2^{+1.0}_{-0.7}$
91 CMS_HWW_boosted_tf_dataResidual_CR2_Bin17
$0.04^{+0.05}_{-0.06}$
92 CMS_HWW_boosted_WJetsCR_mcstat_bin2
93 CMS_res_j_2016APV
94 CMS_HWW_boosted_mu_identification_stat_2016
95 CMS_pileup_2018
96 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin1
97 QCDscale_ttH_ACCEPT_CMS_HWW_boosted
98 CMS_HWW_boosted_WJetsCR_mcstat_bin6
99 CMS_HWW_boosted_btagSFbc_2018
100 CMS_HWW_boosted_mu_identification
101 CMS_VH_HWW_Vqq_WJetsLNu
$1.04^{+0.21}_{-0.18}$
102 CMS_HWW_boosted_WJetsCR_mcstat_bin5
103 CMS_l1_ecal_prefiring_2016APV
104 CMS_HWW_boosted_tf_dataResidual_a_MH_Reco_par1
$0.85^{+0.06}_{-0.06}$
105 CMS_HWW_boosted_jms_2017
106 CMS_HWW_boosted_WJetsCR_mcstat_bin7
107 CMS_HWW_boosted_mu_identification_syst
108 CMS_HWW_boosted_ele_reconstruction
109 CMS_HWW_boosted_wjets_rate
110 CMS_HWW_boosted_mu_identification_stat_2018
111 CMS_HWW_boosted_mu_identification_stat_2016APV
112 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin2
113 ps_fsr_wjets_2016
114 CMS_HWW_boosted_SR2a_mcstat_bin9
115 CMS_HWW_boosted_TopCR_mcstat_bin4
116 CMS_HWW_boosted_ggFpt250to350_mcstat_bin1
117 CMS_HWW_boosted_SR1_fail_mcstat_bin0
118 CMS_HWW_boosted_jmr_2018
119 ps_isr_qqH
120 CMS_HWW_boosted_tf_dataResidual_1b_MH_Reco_par4
$0.92^{+0.15}_{-0.17}$
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
−0.05
0
0.05
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

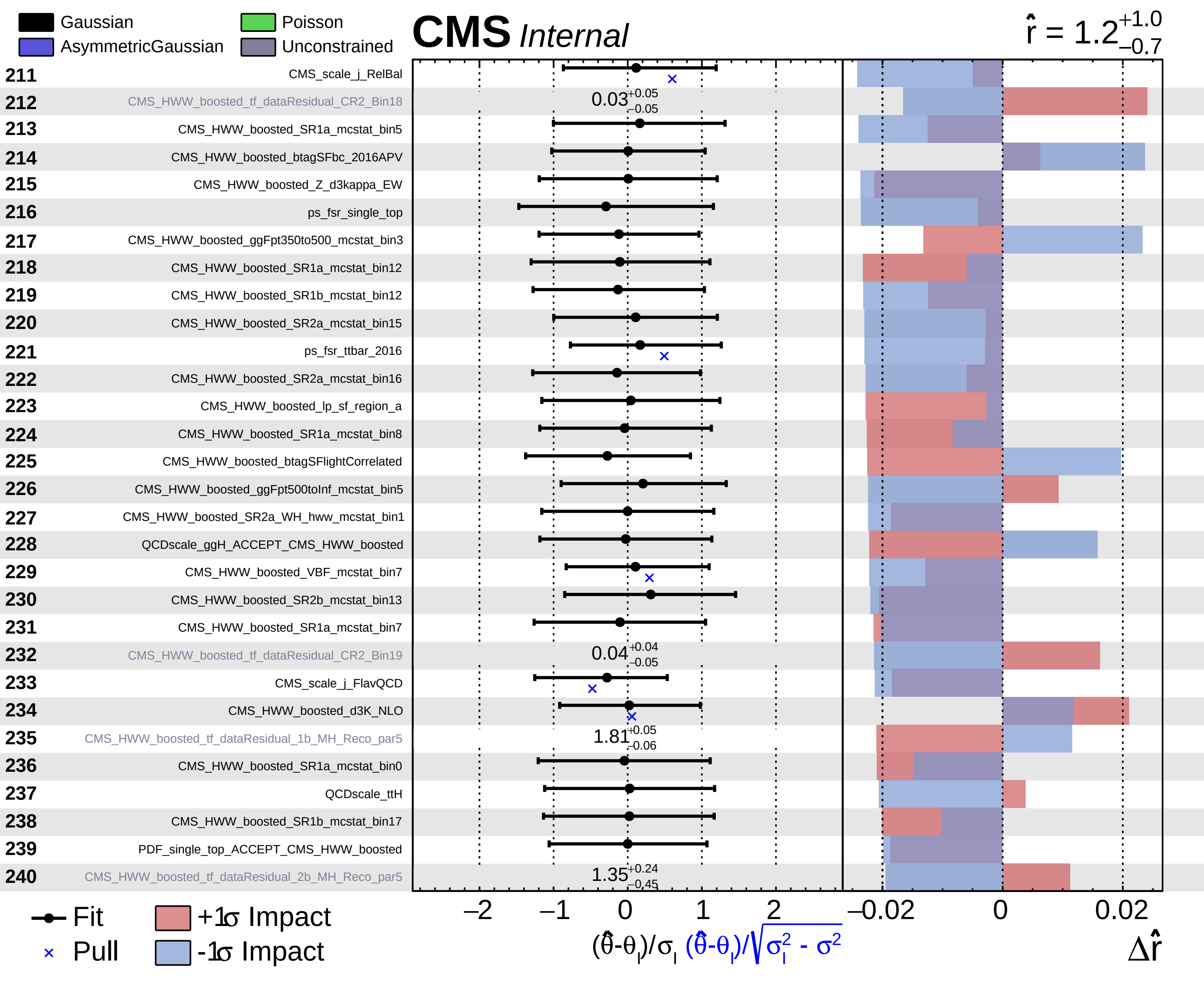


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.2^{+1.0}_{-0.7}$
121 CMS_scale_j_HF_2017
122 CMS_scale_j_RelSample_2016APV
123 CMS_HWW_boosted_tf_dataResidual_1b_MH_Reco_par2
$0.86^{+0.10}_{-0.10}$
124 CMS_HWW_boosted_SR2a_mcstat_bin5
125 CMS_scale_j_RelSample_2017
126 CMS_HWW_boosted_d1K_NLO
127 CMS_HWW_boosted_jmr_2017
128 CMS_HWW_boosted_SR2a_mcstat_bin4
129 ps_isr_ggH
130 CMS_HWW_boosted_TopCR_mcstat_bin1
131 CMS_HWW_boosted_d2K_NLO
132 CMS_scale_j_BBEC1_2016APV
133 QCDscale_rest_bkg
134 CMS_HWW_boosted_tf_dataResidual_1b_MH_Reco_par6
$1.805^{+0}_{-0.023}$
135 CMS_HWW_boosted_ele_reconstruction_2016APV
136 CMS_HWW_boosted_SR2a_mcstat_bin6
137 CMS_HWW_boosted_TopCR_mcstat_bin7
138 CMS_HWW_boosted_ele_trigger_syst_unc
139 CMS_HWW_boosted_tf_dataResidual_1b_MH_Reco_par3
$1.49^{+0.08}_{-0.10}$
140 CMS_HWW_boosted_SR2a_mcstat_bin7
141 ps_fsr_wjets_ggFpt500toInf
142 CMS_HWW_boosted_SR2b_mcstat_bin17
143 ps_fsr_rest_bkg
144 TopCRpass2018_TTbar_mcstat
145 CMS_scale_j_BBEC1
146 CMS_HWW_boosted_ele_identification_2016
147 unclustered_Energy
148 CMS_HWW_boosted_ggFpt350to500_mcstat_bin1
149 CMS_HWW_boosted_ele_trigger_stat_unc_2018
150 CMS_HWW_boosted_ggFpt350to500_mcstat_bin7
−2
−1
0
1
2
−0.04
−0.02
0
0.02
0.04
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

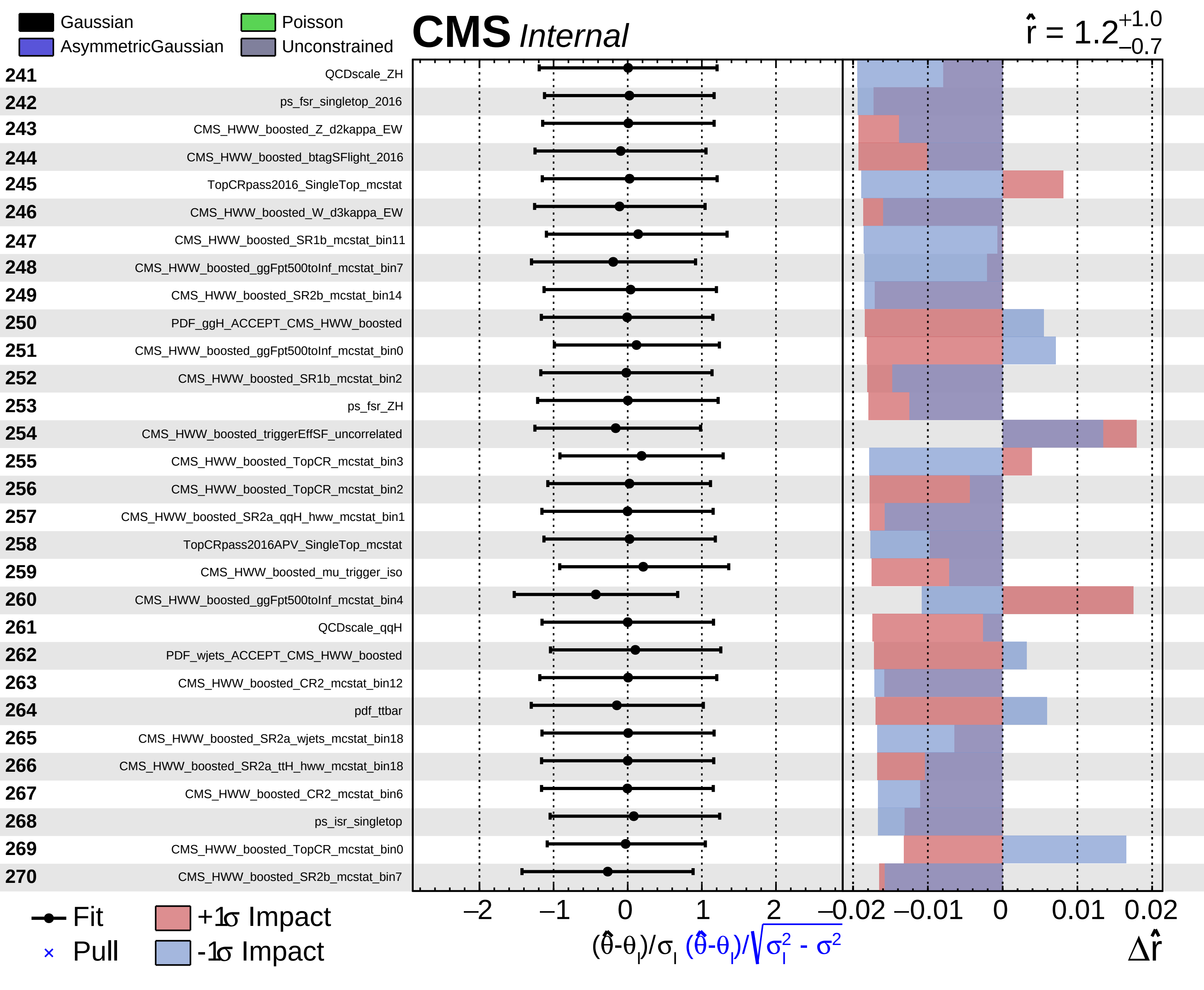

Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.2^{+1.0}_{-0.7}$
151 CMS_HWW_boosted_TopCR_mcstat_bin6
152 CMS_HWW_boosted_ele_trigger_stat_unc_2016APV
153 CMS_HWW_boosted_SR1_fail_mcstat_bin1
154 CMS_HWW_boosted_SR1_pass_mcstat_bin3
155 TopCRpass2016APV_TTbar_mcstat
156 QCDscale_singletop_ACCEPT_CMS_HWW_boosted
157 lumi_13TeV_2017
158 CMS_HWW_boosted_ggFpt350to500_mcstat_bin0
159 CMS_HWW_boosted_SR2b_mcstat_bin8
160 CMS_HWW_boosted_SR1a_mcstat_bin6
161 CMS_pileup_2017
162 CMS_scale_j_Abs
163 CMS_res_j_2018
164 CMS_HWW_boosted_ele_identification
165 CMS_HWW_boosted_SR2a_mcstat_bin10
166 CMS_HWW_boosted_SR2a_mcstat_bin8
167 CMS_HWW_boosted_ele_reconstruction_2018
168 CMS_HWW_boosted_btagSFlight_2017
169 CMS_HWW_boosted_ggFpt250to350_mcstat_bin3
170 CMS_HWW_boosted_VBF_mcstat_bin5
171 CMS_scale_j_RelSample_2018
172 CMS_HWW_boosted_ele_reconstruction_2017
173 CMS_HWW_boosted_tf_dataResidual_2b_MH_Reco_par2
$1.29^{+0.19}_{-0.26}$
174 CMS_HWW_boosted_ggFpt350to500_mcstat_bin6
175 CMS_HWW_boosted_btagSFbcCorrelated
176 ps_fsr_singletop
177 CMS_HWW_boosted_SR1a_mcstat_bin10
178 QCDscale_ggH
179 TopCRpass2017_TTbar_mcstat
180 CMS_HWW_boosted_SR1_pass_mcstat_bin0
−2
−1
0
1
2
−0.04
−0.02
0
0.02
0.04
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

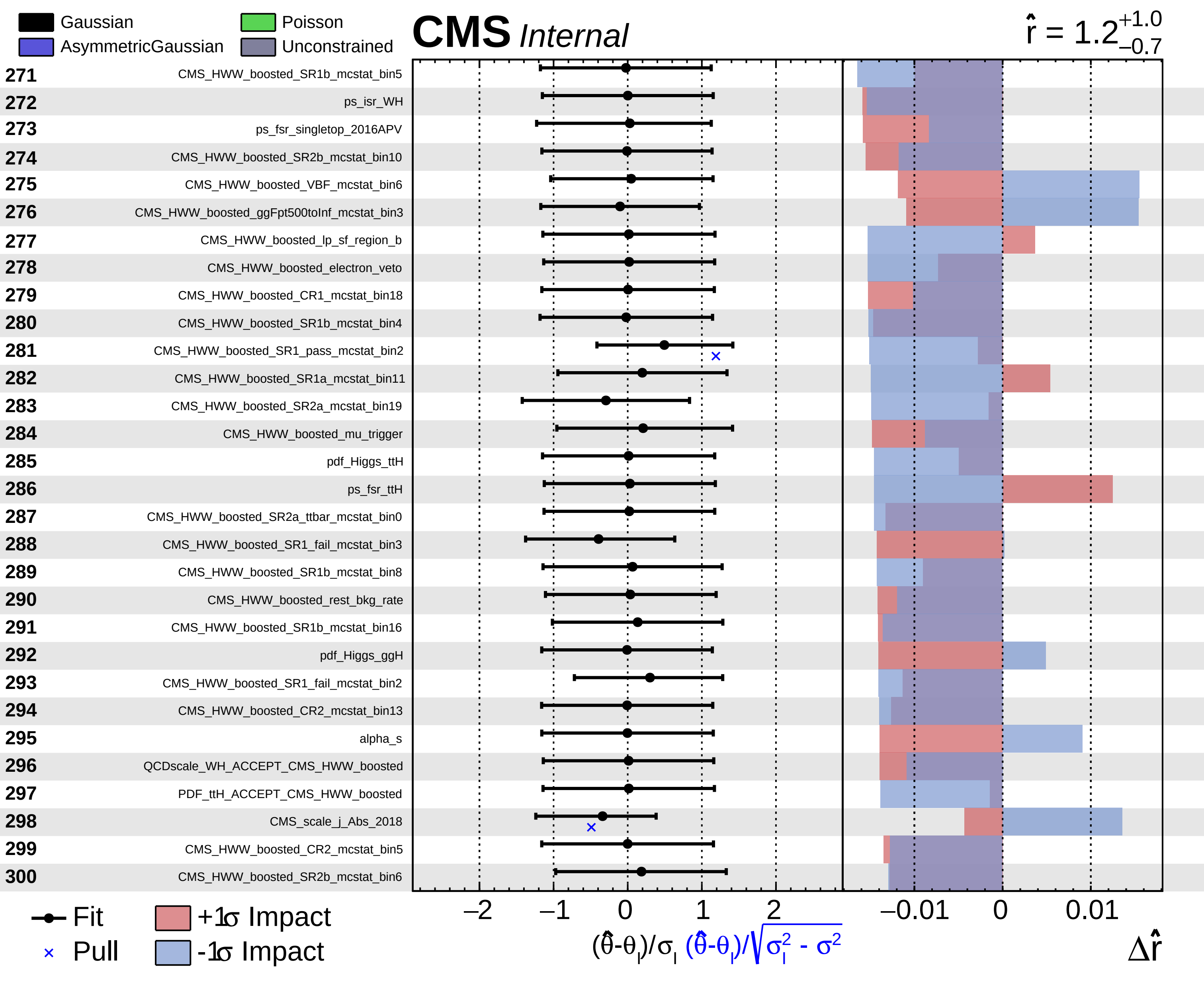


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.2^{+1.0}_{-0.7}$
181 pdf_Higgs_qqH
182 BR_hww
183 QCDscale_ZH_ACCEPT_CMS_HWW_boosted
184 QCDscale_ttbar_ACCEPT_CMS_HWW_boosted
185 CMS_HWW_boosted_ele_reconstruction_2016
186 CMS_HWW_boosted_SR2a_single_top_mcstat_bin18
187 CMS_HWW_boosted_FakeRate_EWK_SF_statistical_uncertainty
188 ps_isr_single_top
189 lumi_13TeV_2018
190 CMS_HWW_boosted_ggFpt250to350_mcstat_bin5
191 CMS_HWW_boosted_SR1b_mcstat_bin0
192 ps_fsr_ttbar_2018
193 CMS_l1_ecal_prefiring_2016
194 pdf_rest_bkg
195 CMS_HWW_boosted_miniisolation_SF_unc
196 CMS_scale_j_HF
197 CMS_HWW_boosted_TopCR_mcstat_bin5
198 CMS_HWW_boosted_ggFpt250to350_mcstat_bin2
199 CMS_HWW_boosted_Vtagger
200 CMS_HWW_boosted_SR1a_mcstat_bin2
201 CMS_HWW_boosted_btagSFlight_2018
202 CMS_HWW_boosted_tf_dataResidual_2b_MH_Reco_par4
$1.77^{+0.25}_{-0.30}$
203 CMS_pileup_2016APV
204 CMS_HWW_boosted_SR1a_mcstat_bin13
205 pdf_single_top
206 CMS_HWW_boosted_mu_isolation
207 CMS_HWW_boosted_ele_identification_2018
208 CMS_pileup_2016
209 CMS_HWW_boosted_ele_identification_2016APV
210 CMS_HWW_boosted_tf_dataResidual_2b_MH_Reco_par6
$1.88^{+0.15}_{-0.13}$
−2
−1
0
1
2
−0.02
0
0.02
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

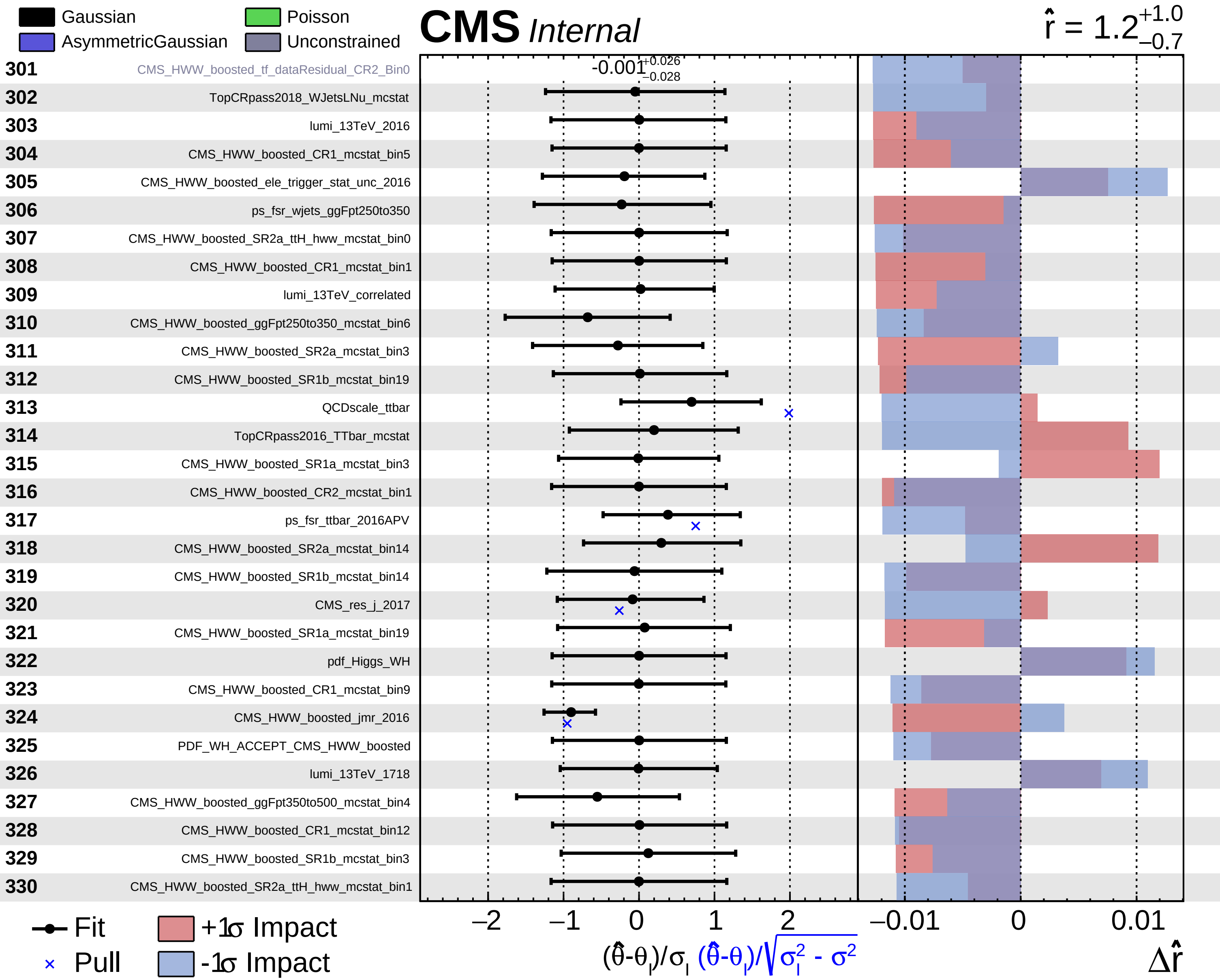


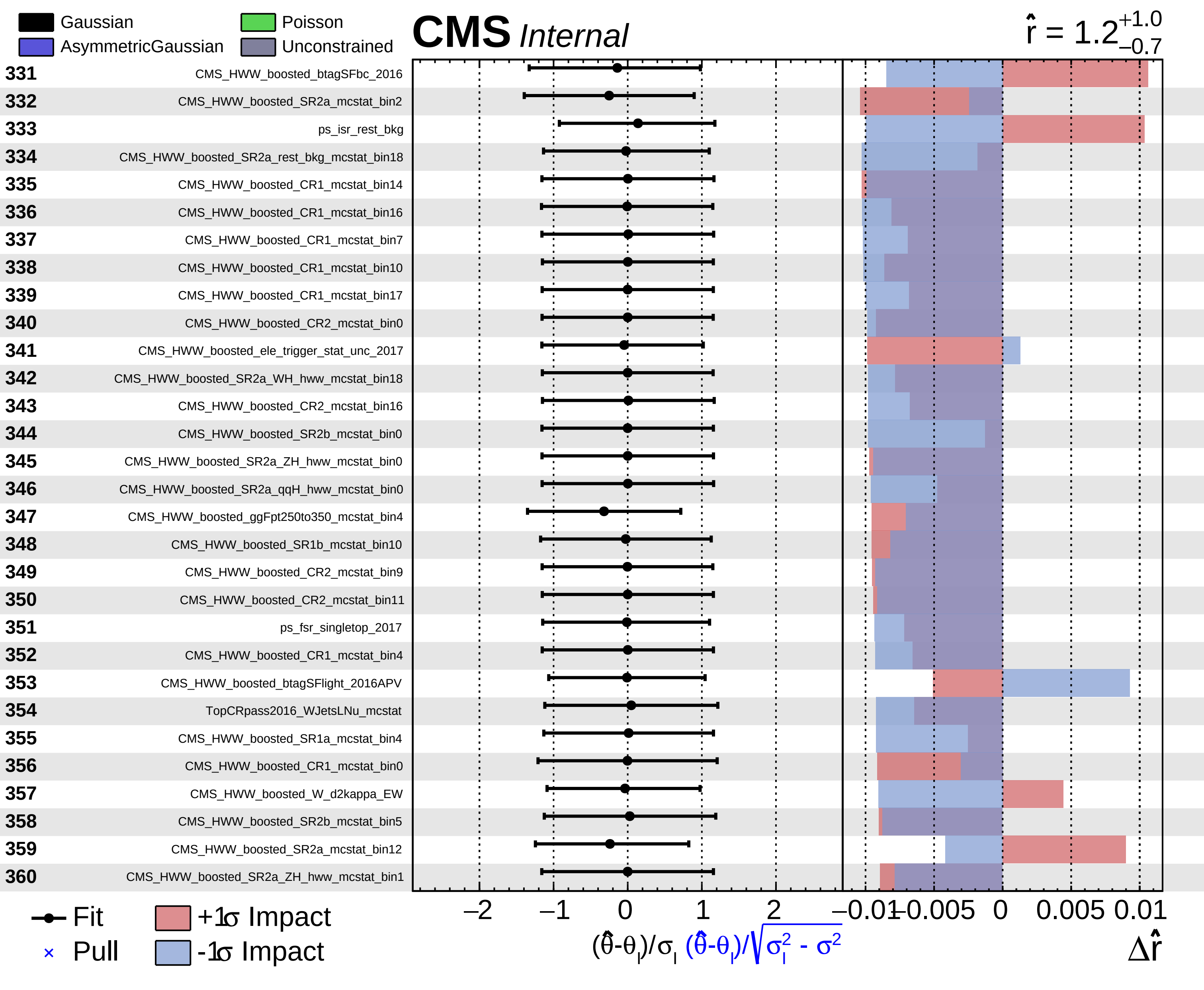
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.2^{+1.0}_{-0.7}$
211 CMS_scale_j_RelBal
212 CMS_HWW_boosted_tf_dataResidual_CR2_Bin18 $0.03^{+0.05}_{-0.05}$
213 CMS_HWW_boosted_SR1a_mcstat_bin5
214 CMS_HWW_boosted_btagSFbc_2016APV
215 CMS_HWW_boosted_Z_d3kappa_EW
216 ps_fsr_single_top
217 CMS_HWW_boosted_ggFpt350to500_mcstat_bin3
218 CMS_HWW_boosted_SR1a_mcstat_bin12
219 CMS_HWW_boosted_SR1b_mcstat_bin12
220 CMS_HWW_boosted_SR2a_mcstat_bin15
221 ps_fsr_ttbar_2016
222 CMS_HWW_boosted_SR2a_mcstat_bin16
223 CMS_HWW_boosted_lp_sf_region_a
224 CMS_HWW_boosted_SR1a_mcstat_bin8
225 CMS_HWW_boosted_btagSFlightCorrelated
226 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin5
227 CMS_HWW_boosted_SR2a_WH_hww_mcstat_bin1
228 QCDscale_ggH_ACCEPT_CMS_HWW_boosted
229 CMS_HWW_boosted_VBF_mcstat_bin7
230 CMS_HWW_boosted_SR2b_mcstat_bin13
231 CMS_HWW_boosted_SR1a_mcstat_bin7
232 CMS_HWW_boosted_tf_dataResidual_CR2_Bin19 $0.04^{+0.04}_{-0.05}$
233 CMS_scale_j_FlavQCD
234 CMS_HWW_boosted_d3K_NLO
235 CMS_HWW_boosted_tf_dataResidual_1b_MH_Reco_par5 $1.81^{+0.05}_{-0.06}$
236 CMS_HWW_boosted_SR1a_mcstat_bin0
237 QCDscale_ttH
238 CMS_HWW_boosted_SR1b_mcstat_bin17
239 PDF_single_top_ACCEPT_CMS_HWW_boosted
240 CMS_HWW_boosted_tf_dataResidual_2b_MH_Reco_par5 $1.35^{+0.24}_{-0.45}$
−2 −1 0 1 2
−0.02 0 0.02
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.2^{+1.0}_{-0.7}$
241 QCDscale_ZH
242 ps_fsr_singletop_2016
243 CMS_HWW_boosted_Z_d2kappa_EW
244 CMS_HWW_boosted_btagSFlight_2016
245 TopCRpass2016_SingleTop_mcstat
246 CMS_HWW_boosted_W_d3kappa_EW
247 CMS_HWW_boosted_SR1b_mcstat_bin11
248 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin7
249 CMS_HWW_boosted_SR2b_mcstat_bin14
250 PDF_ggH_ACCEPT_CMS_HWW_boosted
251 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin0
252 CMS_HWW_boosted_SR1b_mcstat_bin2
253 ps_fsr_ZH
254 CMS_HWW_boosted_triggerEffSF_uncorrelated
255 CMS_HWW_boosted_TopCR_mcstat_bin3
256 CMS_HWW_boosted_TopCR_mcstat_bin2
257 CMS_HWW_boosted_SR2a_qqH_hww_mcstat_bin1
258 TopCRpass2016APV_SingleTop_mcstat
259 CMS_HWW_boosted_mu_trigger_iso
260 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin4
261 QCDscale_qqH
262 PDF_wjets_ACCEPT_CMS_HWW_boosted
263 CMS_HWW_boosted_CR2_mcstat_bin12
264 pdf_ttbar
265 CMS_HWW_boosted_SR2a_wjets_mcstat_bin18
266 CMS_HWW_boosted_SR2a_ttH_hww_mcstat_bin18
267 CMS_HWW_boosted_CR2_mcstat_bin6
268 ps_isr_singletop
269 CMS_HWW_boosted_TopCR_mcstat_bin0
270 CMS_HWW_boosted_SR2b_mcstat_bin7
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

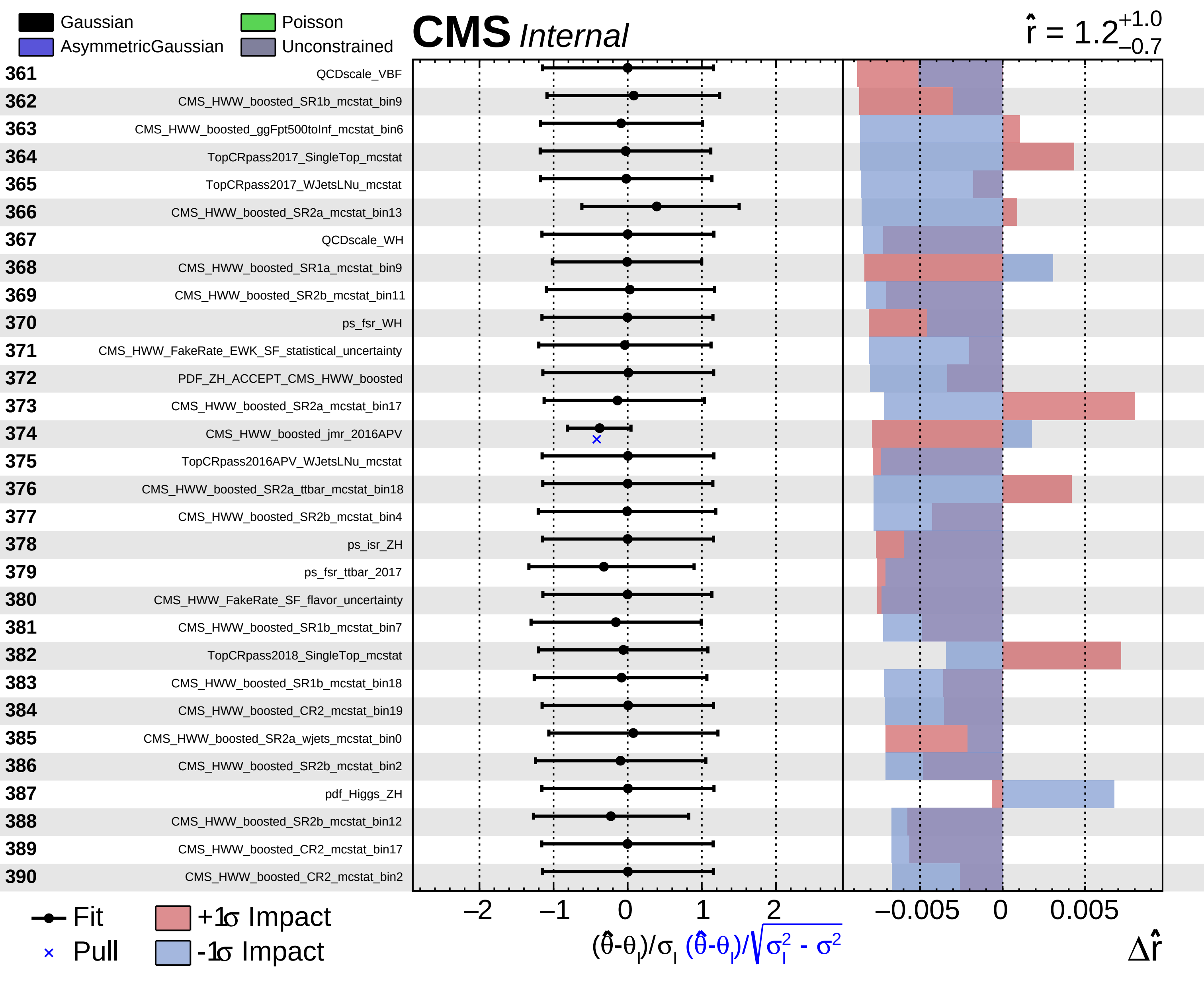


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.2^{+1.0}_{-0.7}$
271 CMS_HWW_boosted_SR1b_mcstat_bin5
272 ps_isr_WH
273 ps_fsr_singletop_2016APV
274 CMS_HWW_boosted_SR2b_mcstat_bin10
275 CMS_HWW_boosted_VBF_mcstat_bin6
276 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin3
277 CMS_HWW_boosted_lp_sf_region_b
278 CMS_HWW_boosted_electron_veto
279 CMS_HWW_boosted_CR1_mcstat_bin18
280 CMS_HWW_boosted_SR1b_mcstat_bin4
281 CMS_HWW_boosted_SR1_pass_mcstat_bin2
282 CMS_HWW_boosted_SR1a_mcstat_bin11
283 CMS_HWW_boosted_SR2a_mcstat_bin19
284 CMS_HWW_boosted_mu_trigger
285 pdf_Higgs_ttH
286 ps_fsr_ttH
287 CMS_HWW_boosted_SR2a_ttbar_mcstat_bin0
288 CMS_HWW_boosted_SR1_fail_mcstat_bin3
289 CMS_HWW_boosted_SR1b_mcstat_bin8
290 CMS_HWW_boosted_rest_bkg_rate
291 CMS_HWW_boosted_SR1b_mcstat_bin16
292 pdf_Higgs_ggH
293 CMS_HWW_boosted_SR1_fail_mcstat_bin2
294 CMS_HWW_boosted_CR2_mcstat_bin13
295 alpha_s
296 QCDscale_WH_ACCEPT_CMS_HWW_boosted
297 PDF_ttH_ACCEPT_CMS_HWW_boosted
298 CMS_scale_j_Abs_2018
299 CMS_HWW_boosted_CR2_mcstat_bin5
300 CMS_HWW_boosted_SR2b_mcstat_bin6
−2
−1
0
1
2
−0.01
0
0.01
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

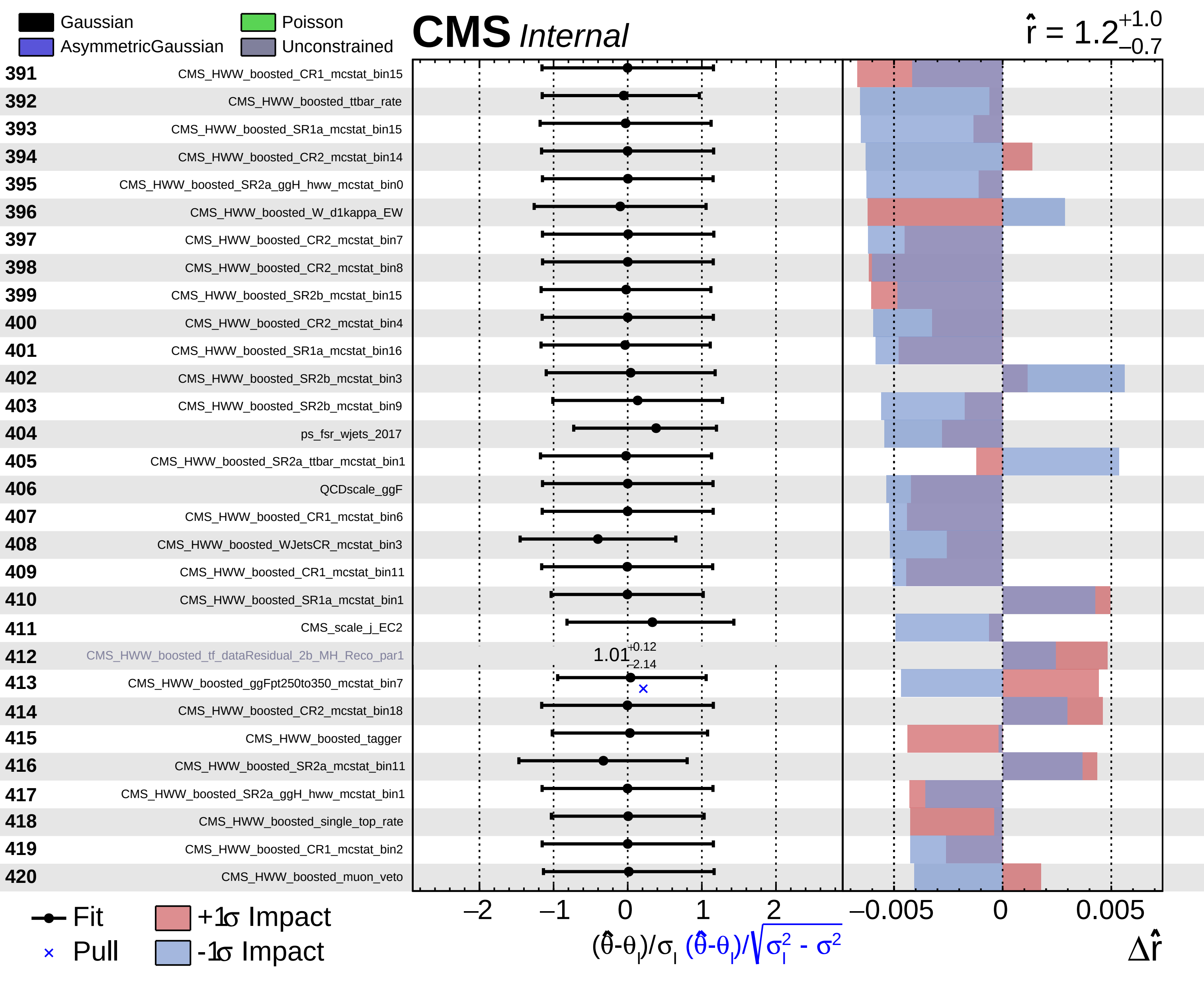

Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.2^{+1.0}_{-0.7}$
$-0.001^{+0.026}_{-0.028}$
301 CMS_HWW_boosted_tf_dataResidual_CR2_Bin0
302 TopCRpass2018_WJetsLNu_mcstat
303 lumi_13TeV_2016
304 CMS_HWW_boosted_CR1_mcstat_bin5
305 CMS_HWW_boosted_ele_trigger_stat_unc_2016
306 ps_fsr_wjets_ggFpt250to350
307 CMS_HWW_boosted_SR2a_ttH_hww_mcstat_bin0
308 CMS_HWW_boosted_CR1_mcstat_bin1
309 lumi_13TeV_correlated
310 CMS_HWW_boosted_ggFpt250to350_mcstat_bin6
311 CMS_HWW_boosted_SR2a_mcstat_bin3
312 CMS_HWW_boosted_SR1b_mcstat_bin19
313 QCDscale_ttbar
314 TopCRpass2016_TTbar_mcstat
315 CMS_HWW_boosted_SR1a_mcstat_bin3
316 CMS_HWW_boosted_CR2_mcstat_bin1
317 ps_fsr_ttbar_2016APV
318 CMS_HWW_boosted_SR2a_mcstat_bin14
319 CMS_HWW_boosted_SR1b_mcstat_bin14
320 CMS_res_j_2017
321 CMS_HWW_boosted_SR1a_mcstat_bin19
322 pdf_Higgs_WH
323 CMS_HWW_boosted_CR1_mcstat_bin9
324 CMS_HWW_boosted_jmr_2016
325 PDF_WH_ACCEPT_CMS_HWW_boosted
326 lumi_13TeV_1718
327 CMS_HWW_boosted_ggFpt350to500_mcstat_bin4
328 CMS_HWW_boosted_CR1_mcstat_bin12
329 CMS_HWW_boosted_SR1b_mcstat_bin3
330 CMS_HWW_boosted_SR2a_ttH_hww_mcstat_bin1
−2
−1
0
1
2
−0.01
0
0.01
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



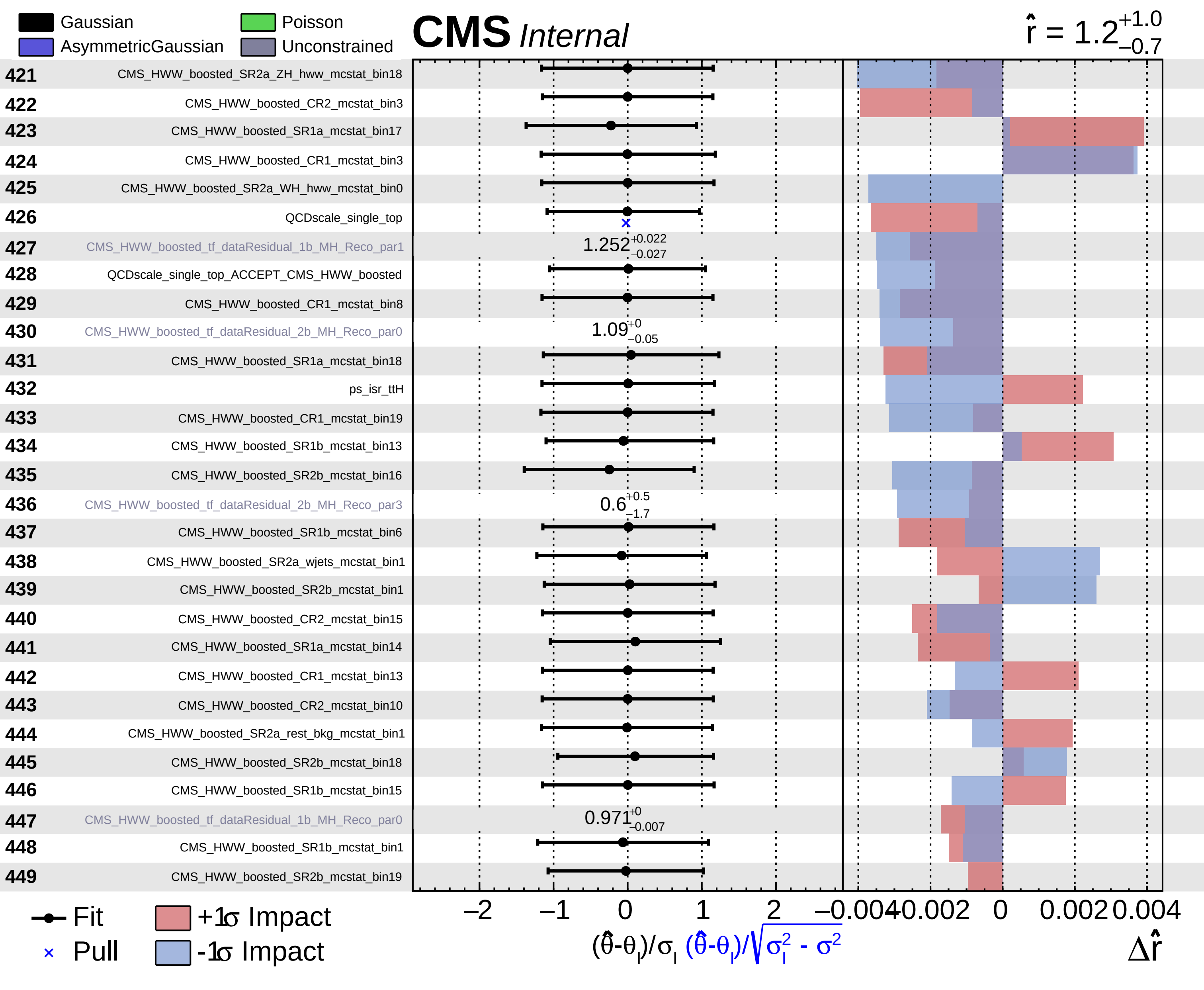
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.2^{+1.0}_{-0.7}$
331 CMS_HWW_boosted_btagSFbc_2016
332 CMS_HWW_boosted_SR2a_mcstat_bin2
333 ps_isr_rest_bkg
334 CMS_HWW_boosted_SR2a_rest_bkg_mcstat_bin18
335 CMS_HWW_boosted_CR1_mcstat_bin14
336 CMS_HWW_boosted_CR1_mcstat_bin16
337 CMS_HWW_boosted_CR1_mcstat_bin7
338 CMS_HWW_boosted_CR1_mcstat_bin10
339 CMS_HWW_boosted_CR1_mcstat_bin17
340 CMS_HWW_boosted_CR2_mcstat_bin0
341 CMS_HWW_boosted_ele_trigger_stat_unc_2017
342 CMS_HWW_boosted_SR2a_WH_hww_mcstat_bin18
343 CMS_HWW_boosted_CR2_mcstat_bin16
344 CMS_HWW_boosted_SR2b_mcstat_bin0
345 CMS_HWW_boosted_SR2a_ZH_hww_mcstat_bin0
346 CMS_HWW_boosted_SR2a_qqH_hww_mcstat_bin0
347 CMS_HWW_boosted_ggFpt250to350_mcstat_bin4
348 CMS_HWW_boosted_SR1b_mcstat_bin10
349 CMS_HWW_boosted_CR2_mcstat_bin9
350 CMS_HWW_boosted_CR2_mcstat_bin11
351 ps_fsr_singletop_2017
352 CMS_HWW_boosted_CR1_mcstat_bin4
353 CMS_HWW_boosted_btagSFlight_2016APV
354 TopCRpass2016_WJetsLNu_mcstat
355 CMS_HWW_boosted_SR1a_mcstat_bin4
356 CMS_HWW_boosted_CR1_mcstat_bin0
357 CMS_HWW_boosted_W_d2kappa_EW
358 CMS_HWW_boosted_SR2b_mcstat_bin5
359 CMS_HWW_boosted_SR2a_mcstat_bin12
360 CMS_HWW_boosted_SR2a_ZH_hww_mcstat_bin1
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.2^{+1.0}_{-0.7}$
361 QCDscale_VBF
362 CMS_HWW_boosted_SR1b_mcstat_bin9
363 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin6
364 TopCRpass2017_SingleTop_mcstat
365 TopCRpass2017_WJetsLNu_mcstat
366 CMS_HWW_boosted_SR2a_mcstat_bin13
367 QCDscale_WH
368 CMS_HWW_boosted_SR1a_mcstat_bin9
369 CMS_HWW_boosted_SR2b_mcstat_bin11
370 ps_fsr_WH
371 CMS_HWW_FakeRate_EWK_SF_statistical_uncertainty
372 PDF_ZH_ACCEPT_CMS_HWW_boosted
373 CMS_HWW_boosted_SR2a_mcstat_bin17
374 CMS_HWW_boosted_jmr_2016APV
375 TopCRpass2016APV_WJetsLNu_mcstat
376 CMS_HWW_boosted_SR2a_ttbar_mcstat_bin18
377 CMS_HWW_boosted_SR2b_mcstat_bin4
378 ps_isr_ZH
379 ps_fsr_ttbar_2017
380 CMS_HWW_FakeRate_SF_flavor_uncertainty
381 CMS_HWW_boosted_SR1b_mcstat_bin7
382 TopCRpass2018_SingleTop_mcstat
383 CMS_HWW_boosted_SR1b_mcstat_bin18
384 CMS_HWW_boosted_CR2_mcstat_bin19
385 CMS_HWW_boosted_SR2a_wjets_mcstat_bin0
386 CMS_HWW_boosted_SR2b_mcstat_bin2
387 pdf_Higgs_ZH
388 CMS_HWW_boosted_SR2b_mcstat_bin12
389 CMS_HWW_boosted_CR2_mcstat_bin17
390 CMS_HWW_boosted_CR2_mcstat_bin2
−2 −1 0 1 2
−0.005 0 0.005
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.2^{+1.0}_{-0.7}$
391 CMS_HWW_boosted_CR1_mcstat_bin15
392 CMS_HWW_boosted_ttbar_rate
393 CMS_HWW_boosted_SR1a_mcstat_bin15
394 CMS_HWW_boosted_CR2_mcstat_bin14
395 CMS_HWW_boosted_SR2a_ggH_hww_mcstat_bin0
396 CMS_HWW_boosted_W_d1kappa_EW
397 CMS_HWW_boosted_CR2_mcstat_bin7
398 CMS_HWW_boosted_CR2_mcstat_bin8
399 CMS_HWW_boosted_SR2b_mcstat_bin15
400 CMS_HWW_boosted_CR2_mcstat_bin4
401 CMS_HWW_boosted_SR1a_mcstat_bin16
402 CMS_HWW_boosted_SR2b_mcstat_bin3
403 CMS_HWW_boosted_SR2b_mcstat_bin9
404 ps_fsr_wjets_2017
405 CMS_HWW_boosted_SR2a_ttbar_mcstat_bin1
406 QCDscale_ggF
407 CMS_HWW_boosted_CR1_mcstat_bin6
408 CMS_HWW_boosted_WJetsCR_mcstat_bin3
409 CMS_HWW_boosted_CR1_mcstat_bin11
410 CMS_HWW_boosted_SR1a_mcstat_bin1
411 CMS_scale_j_EC2
412 CMS_HWW_boosted_tf_dataResidual_2b_MH_Reco_par1
$1.01^{+0.12}_{-2.14}$
413 CMS_HWW_boosted_ggFpt250to350_mcstat_bin7
414 CMS_HWW_boosted_CR2_mcstat_bin18
415 CMS_HWW_boosted_tagger
416 CMS_HWW_boosted_SR2a_mcstat_bin11
417 CMS_HWW_boosted_SR2a_ggH_hww_mcstat_bin1
418 CMS_HWW_boosted_single_top_rate
419 CMS_HWW_boosted_CR1_mcstat_bin2
420 CMS_HWW_boosted_muon_veto
−2 −1 0 1 2
−0.005 0 0.005
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
r̂ = 1.2+1.0−0.7
421 CMS_HWW_boosted_SR2a_ZH_hww_mcstat_bin18
422 CMS_HWW_boosted_CR2_mcstat_bin3
423 CMS_HWW_boosted_SR1a_mcstat_bin17
424 CMS_HWW_boosted_CR1_mcstat_bin3
425 CMS_HWW_boosted_SR2a_WH_hww_mcstat_bin0
426 QCDscale_single_top
427 CMS_HWW_boosted_tf_dataResidual_1b_MH_Reco_par1
1.252+0.022−0.027
428 QCDscale_single_top_ACCEPT_CMS_HWW_boosted
429 CMS_HWW_boosted_CR1_mcstat_bin8
430 CMS_HWW_boosted_tf_dataResidual_2b_MH_Reco_par0
1.09+0−0.05
431 CMS_HWW_boosted_SR1a_mcstat_bin18
432 ps_isr_ttH
433 CMS_HWW_boosted_CR1_mcstat_bin19
434 CMS_HWW_boosted_SR1b_mcstat_bin13
435 CMS_HWW_boosted_SR2b_mcstat_bin16
436 CMS_HWW_boosted_tf_dataResidual_2b_MH_Reco_par3
0.6+0.5−1.7
437 CMS_HWW_boosted_SR1b_mcstat_bin6
438 CMS_HWW_boosted_SR2a_wjets_mcstat_bin1
439 CMS_HWW_boosted_SR2b_mcstat_bin1
440 CMS_HWW_boosted_CR2_mcstat_bin15
441 CMS_HWW_boosted_SR1a_mcstat_bin14
442 CMS_HWW_boosted_CR1_mcstat_bin13
443 CMS_HWW_boosted_CR2_mcstat_bin10
444 CMS_HWW_boosted_SR2a_rest_bkg_mcstat_bin1
445 CMS_HWW_boosted_SR2b_mcstat_bin18
446 CMS_HWW_boosted_SR1b_mcstat_bin15
447 CMS_HWW_boosted_tf_dataResidual_1b_MH_Reco_par0
0.971+0−0.007
448 CMS_HWW_boosted_SR1b_mcstat_bin1
449 CMS_HWW_boosted_SR2b_mcstat_bin19
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Fit
Pull
+1σ Impact
-1σ Impact
(θ̂-θI)/σI (θ̂-θI)/√(σI² - σ²)
Δr̂